Recovering Temporal Regulatory Rewiring: An Interpretable Graph-Based Virtual Perturbation Framework Applied to Ischemic Stroke

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## ■ Paper Summary

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| **Field** | Bioinformatics / Computational Biology — gene-regulatory networks, single-cell transcriptomics, graph neural networks; app |
| **Methodology** | Method/framework paper (computational, no wet-lab). Validated across two mouse cohorts, human GRN convergence, and |
| **Key Contribution** | TSC-GNN recovers edge-level GRN rewiring and master-regulator dynamics from multi-timepoint single-cell transcriptomes |
| **Keywords** | gene-regulatory network; temporal rewiring; virtual perturbation; graph neural network; single-cell transcriptomics; ischemic |
| **Indexing Preference** | Dual-indexed (Scopus + Web of Science); Open Access / Subscription unrestricted |
| **Journals Found** | 31 journals curated and tiered below, drawn from 179 dual-indexed candidates matching the bioinformatics / computational- |

## ■ AMBITIOUS JOURNALS

| # | Journal | Publisher | Index | Access | IF (2024) | CiteScore | Quartile |
|---|---|---|---|---|---|---|---|
| 1 | **Nature Methods** | Nature Research | Scopus + WoS | Subscription | 32.1 | — | Q1 |
| 2 | **Nature Biotechnology** | Nature Research | Scopus + WoS | Subscription | 41.7 | — | Q1 |
| 3 | **Nature Communications** | Nature Research | Scopus + WoS | Open Access | 15.7 | — | Q1 (SJR 4.90) |
| 4 | **Genome Biology** | BioMed Central | Scopus + WoS | Open Access | 9.4 | — | Q1 |
| 5 | **Cell Genomics** | Cell Press | Scopus + WoS | Open Access | 8.4 | — | Q1 |
| 6 | **Cell Systems** | Cell Press | Scopus + WoS | Subscription | 7.70 | — | Q1 |
| 7 | **Nucleic Acids Research** | Oxford University Press | Scopus + WoS | Open Access | 13.1 | — | Q1 |
| 8 | **GigaScience** | Oxford University Press | Scopus + WoS | Open Access | — | — | Q1 (strong OA methods/data) |

***Nature Methods*** *— The flagship venue for new life-science methods; TSC-GNN's edge-level rewiring interpretability and open code with SHA-256 manifests fit its primary-methods mandate, though its GNN-for-transcriptomics scope is more computational than the typical wet-lab method.*

***Nature Biotechnology*** *— Publishes technology/methodology advances with translational reach; the drug-repurposing (HDACi / statins) angle gives a translational hook, but the all-computational, no-validated-assay design is a stretch for this journal's remit.*

***Nature Communications*** *— Broad OA megajournal that regularly hosts computational methods with cross-species and clinical translation; the multi-cohort plus human-blood activation story fits its 'important advance' bar.*

***Genome Biology*** *— A top OA journal for genomics methods and software; the GRN-rewiring framework and reproducible pipeline align tightly with its methods-and-tools scope.*

***Cell Genomics*** *— Cell Press OA genomics journal; the temporal single-cell GRN rewiring in stroke is squarely within its genome-technology-and-biology mandate.*

***Cell Systems*** *— Systems-level life-science methods venue; the graph-based interpretable perturbation framework matches its systems-understanding remit, though it leans more toward wet-lab systems work.*

***Nucleic Acids Research*** *— Home of databases / web servers and computational genomics resources; a strong fit if TSC-GNN is packaged as a usable web server or resource, less so as a pure methods paper.*

***GigaScience*** *— OA big-data methods journal; the fully reproducible conda + SHA-256 pipeline and public-data focus map well to its open-data mission (live IF not retrieved).*

## ■ TARGET JOURNALS

| # | Journal | Publisher | Index | Access | IF (2024) | CiteScore | Quartile |
|---|---|---|---|---|---|---|---|
| 1 | **Bioinformatics** | Oxford University Press | Scopus + WoS | Open Access | 5.4 | — | Q1 |
| 2 | **PLOS Computational Biology** | Public Library of Science | Scopus + WoS | Open Access | 3.5 | — | Q2 |
| 3 | **Briefings in Bioinformatics** | Oxford University Press | Scopus + WoS | Open Access | 7.7 | — | Q1 |
| 4 | **BMC Bioinformatics** | BioMed Central | Scopus + WoS | Open Access | 3.9 | — | Q2 |
| 5 | **Cell Reports Methods** | Cell Press | Scopus + WoS | Open Access | 5.8 | — | Q1/Q2 |
| 6 | **npj Systems Biology and Applications** | Nature Publishing Group | Scopus + WoS | Open Access | 3.55 | — | Q2 |
| 7 | **NAR Genomics and Bioinformatics** | Oxford University Press | Scopus + WoS | Open Access | 2.8 | — | Q2 (JCR) / Q1 (Scopus) |
| 8 | **Bioinformatics Advances** | Oxford University Press | Scopus + WoS | Open Access | 2.6 | — | Q2 |
| 9 | **Genomics, Proteomics and Bioinformatics** | BGI / Oxford University Press | Scopus + WoS | Open Access | 7.2 | — | Q1 |
| 10 | **Molecular Systems Biology** | EMBO Press | Scopus + WoS | Open Access | — | — | Q1 |
| 11 | **IEEE/ACM Transactions on Computational Biology and Bioinformatics** | IEEE / ACM | Scopus + WoS | Subscription | — | — | Q2/Q3 |
| 12 | **Journal of Computational Biology** | Mary Ann Liebert / SAGE | Scopus + WoS | Subscription | — | — | Q3 |

***Bioinformatics*** *— The canonical home for bioinformatics methods; temporal GRN rewiring plus GNN readout is exactly its sweet spot, and the OA option aids visibility.*

***PLOS Computational Biology*** *— OA comp-bio methods journal with a strong single-cell / GNN readership; ideal if you want maximum openness and a methods-friendly review track.*

***Briefings in Bioinformatics*** *— High-IF review-style venue; best if the paper is framed as a methodological review / position piece with the TSC-GNN case study, rather than a pure original-method report.*

***BMC Bioinformatics*** *— Reliable OA methods journal; a pragmatic, high-acceptance home for the algorithmic pipeline and benchmarks.*

***Cell Reports Methods*** *— Cell Press OA methods journal; purpose-built for reproducible computational protocols like the TSC-GNN pipeline and its rewiring benchmark.*

***npj Systems Biology and Applications*** *— Nature-branded OA systems-biology venue; the GRN-rewiring-as-recovery-program framing matches its scope (IF ~3.6).*

***NAR Genomics and Bioinformatics*** *— OUP OA genomics + bioinformatics journal; good fit for the single-cell GRN method, though lower IF (2.8) and JCR Q2.*

***Bioinformatics Advances*** *— OUP OA companion to Bioinformatics for solid-but-narrower methods; a safe, on-scope landing spot (IF 2.6).*

***Genomics, Proteomics and Bioinformatics*** *— BGI / OUP OA journal with rising IF (7.2, Q1); strong fit for the multi-omics GRN rewiring story and China-based authorship.*

***Molecular Systems Biology*** *— EMBO OA systems-biology flagship; excellent fit for network-rewiring biology if the biological (not just computational) narrative is strengthened (live IF not retrieved).*

***IEEE/ACM Transactions on Computational Biology and Bioinformatics*** *— Society venue for algorithmic / statistical comp-bio; fits the GNN method core but is lower-prestige than the biology-native options (live IF not retrieved).*

***Journal of Computational Biology*** *— Long-standing comp-bio methods journal; a dependable, lower-tier home for the algorithmic contribution (live IF not retrieved).*

## ■ SAFE JOURNALS

| # | Journal | Publisher | Index | Access | IF (2024) | CiteScore | Quartile |
|---|---|---|---|---|---|---|---|
| 1 | **Frontiers in Bioinformatics** | Frontiers Media | Scopus + WoS | Open Access | 3.6 | — | Q2 |
| 2 | **BMC Genomics** | BioMed Central | Scopus + WoS | Open Access | 3.9 | — | Q1/Q2 |
| 3 | **BioData Mining** | BioMed Central | Scopus + WoS | Open Access | 6.1 | — | Q2 |
| 4 | **Scientific Reports** | Nature Research | Scopus + WoS | Open Access | 3.8 | — | Q1 (multidisciplinary) |
| 5 | **Computers in Biology and Medicine** | Elsevier | Scopus + WoS | Subscription | 7.0 | — | Q1 (■ On Hold by WoS 2024) |
| 6 | **Human Genomics** | BioMed Central | Scopus + WoS | Open Access | 4.3 | — | Q1 |
| 7 | **Frontiers in Computational Neuroscience** | Frontiers Media | Scopus + WoS | Open Access | 2.3 | — | Q3 |
| 8 | **Frontiers in Neuroinformatics** | Frontiers Media | Scopus + WoS | Open Access | 2.5 | — | Q3 |
| 9 | **Algorithms for Molecular Biology** | BioMed Central | Scopus + WoS | Open Access | 1.7 | — | Q3/Q4 |
| 10 | **Computational Biology and Chemistry** | Elsevier | Scopus + WoS | Subscription | 3.4 | — | Q2/Q3 |
| 11 | **Journal of Integrative Bioinformatics** | De Gruyter | Scopus + WoS | Open Access | 2.0 | — | Q4 |

***Frontiers in Bioinformatics*** *— OA frontier journal, fast and broad; a low-risk home for the method with decent visibility (IF 3.6).*

***BMC Genomics*** *— Broad OA genomics journal; suitable if the paper is angled toward the stroke transcriptomics findings rather than the method.*

***BioData Mining*** *— BMC OA journal for biomedical ML / data-mining; fits the ML + single-cell angle (IF 6.1, higher than typical Safe).*

- ***Scientific Reports*** *— Nature Portfolio mega-journal, very high acceptance; a dependable safety net for a complete, reproducible method (IF 3.8).*
- ***Computers in Biology and Medicine*** *— Elsevier journal for biomedical ML; good topical fit but was placed On Hold by WoS in 2024 — verify its current indexing status before submitting (IF ~7.0).*
- ***Human Genomics*** *— OA genomics journal; relevant if the human-stroke-blood activation result is foregrounded (IF 4.3).*
- ***Frontiers in Computational Neuroscience*** *— OA neuroscience-methods journal; fits the stroke / rewiring application angle, lower IF (2.3).*
- ***Frontiers in Neuroinformatics*** *— OA neuroinformatics venue; appropriate if the software / tooling aspect is emphasized (IF 2.5).*
- ***Algorithms for Molecular Biology*** *— BMC OA algorithms journal; a niche but on-scope home for the graph method (IF 1.7).*
- ***Computational Biology and Chemistry*** *— Elsevier journal for computational life sciences; acceptable fit, moderate tier (IF 3.4).*
- ***Journal of Integrative Bioinformatics*** *— OA integrative-bioinformatics journal; a modest, near-guaranteed-acceptance fallback (IF ~2.0).*

## ■ Paper Quality Assessment

The paper was evaluated on five dimensions (scored 1–5) to determine the appropriate tier anchor for recommendations.

| Dimension | Score | Visual | Assessment |
|---|---|---|---|
| **Novelty** | 4 / 5 | ■■■■□ | TSC-GNN is a novel synthesis of state-conditioned GRNs, GNN message-passing, and edge-level rewiring interpretability; it extends (rather than invents) each component but introduces a distinct recovery-oriented virtual-perturbation framing. |
| **Methodological Rigor** | 4 / 5 | ■■■■□ | Validated across two mouse cohorts, human GRN convergence, and three independent perturbation modalities with explicit permutation testing; limited only by the absence of wet-lab confirmation. |
| **Contribution Breadth** | 3 / 5 | ■■■□□ | Advances the interpretable virtual-perturbation sub-field applied to stroke recovery; impact is concentrated rather than field-wide. |
| **Evidence Quality** | 4 / 5 | ■■■■□ | Longitudinal 24 h to 14 d design, cross-cohort reproducibility (rho = 0.48-0.55), and cross-species triangulation give robust computational evidence, though entirely in silico. |
| **Clarity of Advance** | 4 / 5 | ■■■■□ | The manuscript openly reports that a fixed graph plus linear readout does not beat linear baselines (0/90), reframing the contribution as interpretability - a clear, honest benchmark. |

**Total Score: 19 / 25**

Best realistic fit is the Target tier. Ambitious journals are aspirational reaches: the central contribution is interpretability rather than accuracy gains, which top methods venues weigh heavily. Lead with Bioinformatics (Oxford), Cell Reports Methods, or PLOS Computational Biology.